

Dart Emerald Performance PC

A decent performer

The Dart Emerald Performance PC was more or less similar to the Value PC. It has a whopping 256 MB RAM, a Seagate Barracuda 7200 RPM 40 GB hard drive and a Pentium 4 2.4 GHz on an Intel 845GVAD2 motherboard that provides no upgrade options when it comes to graphics, as there is no AGP slot. No onboard LAN is provided either. Also, included is a CD-ROM drive and a 17-inch monitor. The multimedia keyboard and the optical mouse complement the package well. However, there weren't any speakers. It comes bundled with Windows XP Home edition, but misses out on office suites and games.



The performance gain over the Value PC is hardly noticeable. In the gaming arena, it barely showed any significant improvement over the value PC, with the variation being a little over 1 per cent.

Quake III saw an increase of a mere 4 frames, and the *UT2003* results were similar. However, the ZDbench content creation saw its score jump by a whopping 8 points—a good example of what the extra 128 MB can do to your system. Similarly, the PSbench test results saw the time reduce drastically by 133 seconds over the Value PC.

Priced at Rs 39,000, it is definitely expensive. Also, increasing the RAM to 256 MB, including a 17-inch monitor and cramping in a CD-ROM drive does not make this system a Performance PC, since important sub-systems such as the graphics are on the lower side. It also lacks upgradeability.

Price: Rs 39,000
+ Good hardware configuration
- No speakers

Dart Emerald Performance PC		B-	
Performance	▬ ▬ ▬ ▬ ▬		
Features	▬ ▬ ▬ ▬ ▬		
Build quality	▬ ▬ ▬ ▬ ▬		
Value for money	▬ ▬ ▬ ▬ ▬		

Datamini Festiva MG

A Vanilla PC

The Festiva MG model gave the impression of being a standard run-of-the-mill type PC, with no frills or thrills thrown at you. Design-wise, it lacked the aesthetic appeal associated with a good branded PC, and looks quite dated.

On the configuration front, the PC did come up with decent components—a Pentium 4 2.4 GHz with 256 MB DDR 266 RAM plugged on an MSI 845 chipset-based motherboard. Upgradeability was limited by the absence of an AGP slot and an onboard LAN controller. The monitor displayed below-average quality and seemed like a 14-inch, though it was a 15-inch. The multimedia keyboard and the ball mouse were not up to the mark, when compared to those of the other PCs.

In order to lower the price tag, critical components such as the internal modem and speakers did not make it to our lab; nei-

ther is an OS nor extra software or games included—all you can expect is the drivers CD.

The Festiva MG could barely beat the similarly configured Dart Performance PC. It was on par with it in the processor sub-system tests, as well as all the processor tests. However, the gaming tests saw the Dart PC shoot past by a small margin.

The Datamini Festiva MG offers a decent package at Rs 31,700, but sans an OS and other software. The Dart Performance PC beats it hands down, but costs an extra Rs 8,000, which includes the cost of the OS and 3-years warranty. Bottom line—not a good buy.

Price: Rs 31,700
+ None
- Non-upgradeable

Datamini Festiva MG		B	
Performance	▬ ▬ ▬ ▬ ▬		
Features	▬ ▬ ▬ ▬ ▬		
Build quality	▬ ▬ ▬ ▬ ▬		
Value for money	▬ ▬ ▬ ▬ ▬		

64-bit Computing

The launch of 64-bit processors from AMD and Apple heralds a new era in desktop computing. HCL and Zenith have already started work in this direction. So what do all the extra 32-bits mean to you? Will the new processor be faster than what we currently have on our desktop? To put it right, it's no more about the clock cycle speed, but rather about the amount of data that is processed in a single clock cycle. A 64-bit processor will process double the amount of data, as compared to any 32-bit CPU. This translates to improved overall system performance in demanding applications such as game development studios and video encoding, where the file sizes run into gigabytes. Another advantage is the increase in the memory addressable capacity, which is currently at 4 GB due to the 32-bit addressable lines. With 64-bit, you can address a mind-boggling 16 billion GB of memory.

AMD has launched two variants of its 64-bit desktop processor—the Athlon 64 and the Athlon 64 FX, with subtle differences between them. The latter is a better option and offers a dual-channel memory controller. Also, both the variants come with 128 KB of L1 cache and 1 MB of L2 cache, with support for AMD's 3DNow! and SSE2 instructions.

Similarly, Apple has come up with its G5 processor for Mac users. Though Intel has the 64-bit Itanium processor for the server market, no word is out from them on the desktop front. Currently though, 64-bit OS, applications and device drivers are not available off the shelf. The new Athlons will work as 32-bit processors in Windows XP, their full potential coming into the picture once Microsoft launches its much awaited 64-bit version of Windows. Apple's latest OS, Panther, will get an upgrade that will support 32-bit applications making 64-bit requests to the CPU, but a true 64-bit OS has not been formally announced.